

ASSIGNMENT 2

Course Name : Human Computer Interaction

Course Code : ☐ DMT212 ☐ SWE203

Lecturer : Longbiao CHEN

Student ID : _____

Full Name : _____

Submission Date : _____

Feedback from Lecturer:	Mark:

Own Work Declaration

I hereby understand that my work would be checked for plagiarism or other misconduct, and the softcopy would be saved for future comparison.

I hereby confirm that all the reference or source of citation have been correctly listed or presented and I clearly understand the serious consequence caused by any intentional or unintentional misconduct.

This work is not made on any work of other students (past or present), and it has not been submitted to any other courses or institutes before.

Signature:

Date:

Total Marks: 10 **Estimated Time: 60 minutes**

Question 1 (4 marks)

A *T9 predictive* text input method (Figure Q1(a)) for entering text on mobile devices is proposed in comparison with the *conventional* text input method (Figure Q1(b)) using a *qwerty* keyboard. In order to answer the following research question:

Is the T9 Predictive text input method more efficient than the conventional method?

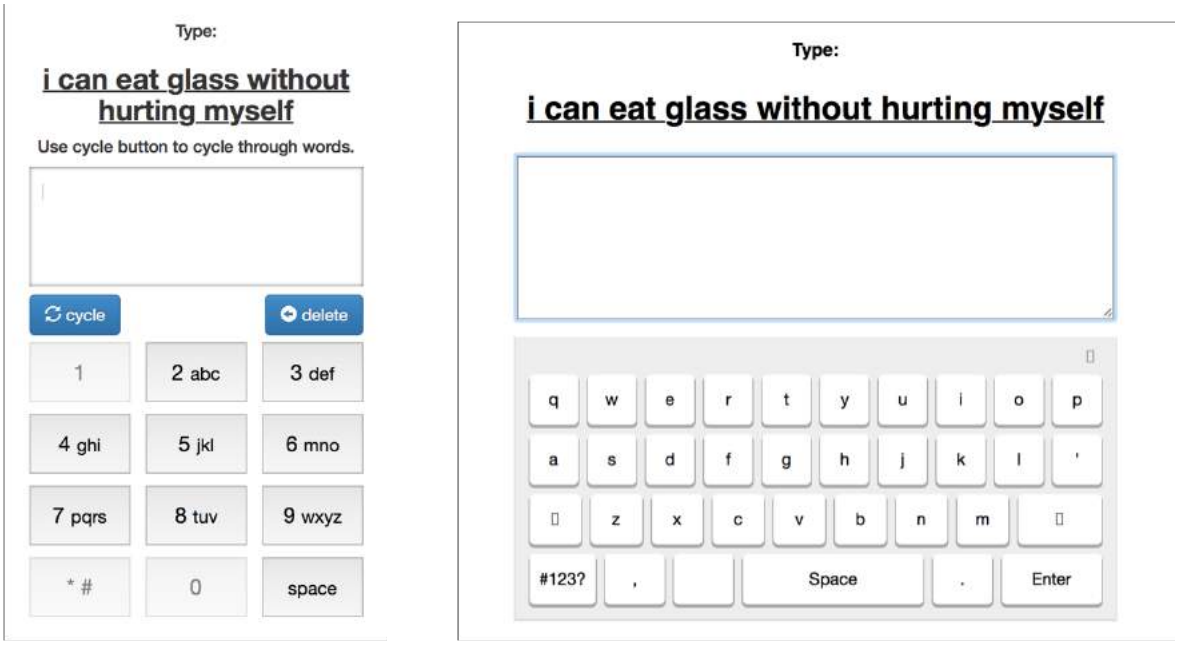
You are asked to conduct an HCI experiment. Follow the instructions below.

1. Recruit 8 students from your university.
2. Setup a mobile phone web browser, and point to the following two pages respectively:

   T9 Predictive: http://aws.longbiaochen.com/hci/t9/

   Conventional: http://aws.longbiaochen.com/hci/qwerty/

3. For each student, ask them to type in the phrase as indicated in the web page using the T9 predictive method, and then using the conventional method. Record the typing time for each task with a watch (in seconds). Collect the initial, sex, and age of the student via a questionnaire.



(a) T9 Predictive Text Input Method (b) Conventional Text Input Method

Figure Q1

Answer the following questions.

(a) Name the independent variable in this experiment design. [1 mark]

(b) Name the dependent variable in this experiment design. [1 mark]

(c) List at least two random variables in this experiment design. [2 marks]

Question 2 (6 marks)

Based on the HCI experiment described in Question 1, the data about typing time (measured in seconds per task) are collected in Table Q2(a). Specifically, the *T9* and *Conventional* columns correspond to the typing time with the 8 participants using the corresponding typing methods. In order to test the hypothesis, an *F-Test* is conducted over the dataset. The test results are listed in Table Q2(b). A threshold for *the p value* is set to 0.05.

Table Q2(a)

Participant	*T9*	*Conventional*
P1		
P2		
P3		
P4		
P5		
P6		
P7		
P8		

Table Q2(b)

F-Test Two-Sample for Variances		
	T9	*Conventional*
Mean		
Variance		
Observations		
df		
F		
P(F<=f) one-tail		
F Critical one-tail		

Figure Q5

Answer the following questions.

(a) Fill in Table Q2(a) with your real-world experimental data. [1 mark]

(b) Write the null hypothesis in this experiment. [2 marks]

(c) Fill in Table Q2(b) with your *F-Test* results, determine and explain whether the null hypothesis is accepted or rejected. [3 marks]

Hint: F-test can be conducted with Microsoft Excel Analysis ToolPak. Visit:

https://www.excel-easy.com/data-analysis/analysis-toolpak.html

https://www.excel-easy.com/examples/f-test.html

for tutorials and guides.

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